Submission PY1-FINAL-SUB06

Question PY1-FINAL-Q01
notation changes halfway:
1. range(6) gives 0 through 5
2. keep even i: 0, 2, 4
3. squares ≈ > [0, 4, 16]
marks, subscripts, and units are mixed up
result written as: [0, 4, 16] (please interpret)

Question PY1-FINAL-Q02
notation changes halfway:
1. counts ≈ Counter(xs)
2. for x in xs: if counts[x] ≈ ≈ 1: return x
3. return None; two linear passes ⇒ O(n) time, O(n) space
marks, subscripts, and units are mixed up
result written as: scan counts, then scan in order; return first count==1 else None; O(n) time, O(n) space (please interpret)

Question PY1-FINAL-Q03
notation changes halfway:
1. A single pass fixes only adjacent inversions encountered once.
2. Repeat until no swaps (or n–1 passes), shrinking the suffix after each pass.
marks, subscripts, and units are mixed up
result written as: repeat passes or track the shrinking unsorted suffix; one pass alone is not a full sort (please interpret)

Question PY1-FINAL-Q04
notation changes halfway:
1. linear scan: up to 64 comparisons, Θ(n)
2. binary search: at most ■log■(64)■+1 comparisons, Θ(log n)
marks, subscripts, and units are mixed up
result written as: linear Θ(n), up to 64; binary Θ(log n), about 7 (please interpret)

Question PY1-FINAL-Q05
notation changes halfway:
1. Removing shifts later elements left while the iterator advances.
2. Build a filtered list or iterate over a copy, then replace the original.
marks, subscripts, and units are mixed up
result written as: mutation shifts elements and skips checks; use a filtered copy or iterate over a copy (please interpret)

Question PY1-FINAL-Q06
notation changes halfway:

1. range(5) gives 0 through 4
2. keep even i: 0, 2, 4
3. squares ≈ > [0, 4, 16]

marks, subscripts, and units are mixed up

result written as: [0, 4, 16] (please interpret)

Question PY1-FINAL-Q07

notation changes halfway:

1. counts ≈ Counter(xs)
2. for x in xs: if counts[x] ≈ ≈ 1: return x
3. return None; two linear passes ⇒ O(n) time, O(n) space

marks, subscripts, and units are mixed up

result written as: scan counts, then scan in order; return first count==1 else None; O(n) time, O(n) space (please interpret)

Question PY1-FINAL-Q08

notation changes halfway:

1. A single pass fixes only adjacent inversions encountered once.
2. Repeat until no swaps (or n–1 passes), shrinking the suffix after each pass.

marks, subscripts, and units are mixed up

result written as: repeat passes or track the shrinking unsorted suffix; one pass alone is not a full sort (please interpret)

Question PY1-FINAL-Q09

notation changes halfway:

1. linear scan: up to 64 comparisons, Θ(n)
2. binary search: at most ■log■(64)■+1 comparisons, Θ(log n)

marks, subscripts, and units are mixed up

result written as: linear Θ(n), up to 64; binary Θ(log n), about 7 (please interpret)

Question PY1-FINAL-Q10

notation changes halfway:

1. Removing shifts later elements left while the iterator advances.
2. Build a filtered list or iterate over a copy, then replace the original.

marks, subscripts, and units are mixed up

result written as: mutation shifts elements and skips checks; use a filtered copy or iterate over a copy (please interpret)